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Determining quality of water in reservoir using machine learning

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ABSTRACT

Water quality is one of the most critical issues in reservoir management owing to its strong effects on the natural environment and human life. This study establishes a machine learning approach for predicting Carlson's Trophic State Index, which is a frequently used metric of water quality in reservoirs. Data collected over ten years (1995–2016) from the stations at 20 reservoirs in Taiwan were preprocessed as the input for the modeling system. Four well-known artificial intelligence techniques, artificial neural networks (ANNs), support vector machines, classification and regression trees, and linear regression, were used to analyze in baseline and ensemble scenarios. A user-friendly interface that integrates a metaheuristic regression model was developed to evaluate the predictive performance, and to compare it with those in the two constituent scenarios. The comprehensive comparison demonstrated that the ensemble ANNs model, based on a tiering method, is more accurate than the other single, ensemble models and hybrid metaheuristic regression model. Both the accuracy of prediction and the efficacy of application are considered to support practitioners in planning water management works. Accordingly, this study provides a novel approach for potential use in water quality assessment.

1. Introduction

Water in reservoirs is a vital resource for all living organisms. Reservoirs provide not only pure water and habitats for a complex variety of aquatic life, but also water for such diverse purposes as agriculture, industry, and everyday human consumption. Various properties of water in reservoirs, especially its quality, must be assessed. Assessing the quality of water critically enables managers to develop optimal water resources management plans.

Many conventional studies have used a wide range of water quality indexes in relation to various bodies of water such as lakes, rivers, and dam reservoirs (Parparov et al., 2014; Sofwan and Fong, 2014; Walker et al., 2015). Carlson's Trophic State Index (CTSI), a fundamental index that was developed in 1977 (Carlson, 1977), is commonly used by water management agencies and organizations around the world. CTSI is a particularly popular and useful index of water quality, and numerous studies have used it as the primary metric of water quality (Prasad and Siddaraju, 2012; Taheri Tizro et al., 2016).

The Taiwan Environmental Protection Administration (TEPA) has adopted CTSI as the official index of water quality, accounting for its state of eutrophication, in reservoirs (Table 1). The state of eutrophication is specified as: oligotrophic, mesotrophic or eutrophic (Taiwan Environmental Protection Administration, 2017b). CTSI is calculated from three independent water quality variables, which are: chlorophyll-a (Chl-a) concentration, total phosphorus (TP) concentration and Secchi depth (SD).

Measuring the concentration of chlorophyll-a and total phosphorus requires sophisticated techniques that involve chemical experiments, equations, specific analyses of water samples. In contrast, Secchi depth can be measured manually in the reservoirs without chemical experiments or modern technology but it is probably the most unstable parameter because of its strong dependence on weather (including season, turbidity and other facts) (Chang and Liu, 2015). Consequently, alternative variables or methods for accurately measuring CTSI would represent a significant development in the field.

While classical methods can effectively evaluate the Carlson's Trophic State Index by assessing the above parameters (Baban, 1996; Barki and Singa, 2014), recent researchers have sought more active solutions for water management. Several studies have developed diverse water quality models based on spatial variations and environmental parameters using multiple linear regression methods (Slaughter et al., 2017; Tomas et al., 2017; Wu et al., 2018). In particular, artificial intelligence (AI) based approaches, which have been typically used for the similar purpose in recent years, have advantages over the traditional techniques as they take into account nonlinear relationships among the influential factors and reduce the necessary sophistication of deriving experimental equations.

Various AI based approaches have been confirmed to be useful in

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Table 1
Degree of eutrophication level based on CTSI.

CTSI index value	Eutrophication level
CTSI < 40	Oligotrophic
40 ≦ CTSI ≦ 50	Mesotrophic
CTSI > 50	Eutrophic

solving water quality management problems. For instance, Antanasijević et al. (2014) proposed a general regression neural network (GRNN) model for the prediction of dissolved oxygen (DO) in the Danube River and then compared the model with Monte Carlo simulation result (Antanasijević et al., 2014). They concluded that the GRNN could effectively forecast the DO concentration. Chau (2006) reviewed the progress on the water quality modeling and concluded that integrating AI technologies, including knowledge-based systems, genetic algorithm, ANNs, and fuzzy inference system, with domain knowledge into numerical modeling systems have been proven effective (Chau, 2006).

Moreover, Khan and See (2016) developed a water quality prediction model associated with water quality factors using ANNs and time-series analysis (Khan and See, 2016). Liou and Lo (2005) proposed an index model for quality evaluation of reservoir based on the formula of similarity membership functions in the fuzzy c-means clustering algorithm (Liou and Lo, 2005). Najah et al. (2009) attempted to predict water quality parameters at Johor River Basin utilizing an ANNs technique. Their results show that the proposed ANNs model has a great potential to predict the water quality parameters with Mean Absolute Percentage Error of 10% (Najah et al., 2009).

AI techniques may involve complex manual implementation that can reduce their practical efficiency for water quality managers. Practitioners need to be trained in programming or coding to apply such complicated models. To improve predictive accuracy and the further applicability of AI based models, a comprehensive comparison of the applications of AI based models using various software package must be carried out. Although many data analysis platforms do not require extensive manipulation in the application of various AI models, most support only basic techniques without optimization.

This work further develops a user-friendly interface in MATLAB for the use by practitioners who do not have a background in programming. The proposed interface is based on a nature-inspired metaheuristic regression system, which integrates an optimization algorithm, called the firefly colony, and a variation of the support vector machine, that is least squares support vector regression. The quality of water is also predicted by four basic AI techniques, which involves: ANNs, support vector machines for regression (SVR), Classification and Regression Trees (CART) and linear regression (LR), using four data analytic platforms (RapidMiner Studio, Microsoft Azure Machine Learning Studio, WEKA, and IBM SPSS Modeler).

The predictive AI system herein is constructed with four single AI models, four ensembles (*i.e.*, voting, bagging, stacking, tiering), and a hybrid metaheuristic regression. The baseline models include four individual models using four AI techniques: ANNs, SVR, CART and LR, respectively. Then, four ensemble models, which are created by combining the listed single models, use voting, bagging, stacking, and tiering methods. Finally, three metaheuristic components are used to enhance the conventional firefly algorithm (FA). It is integrated with least squares support vector regression (LSSVR) to generate the AI hybrid metaheuristic model.

The objective of this research is to develop a versatile water quality modeling approach, including single, ensemble, and hybrid models. The analytical procedure has two parallel targets: predictive accuracy and applicability of models. The system will enable managers or water scientist who use AI for water management to choose the best analytical tools for various purposes.

2. Literature review

Numerous indicators and indices of trophic status and quality of aquatic ecosystems are available (Liou et al., 2004; Parparov et al., 2014). Some conventional studies use the particular concentrations in water sample to evaluate water quality directly while others are based on composite indexes that involve many parameters.

Matthews et al. (2002) utilized an 11 year data-set of numerous water quality parameters: total phosphorus concentration (TP), chlorophyll-a concentration (Chl-a), Secchi depth (SD), total nitrogen concentration (TN), and dissolved inorganic nitrogen concentration to assess trophic state and nutrient limitation of water in Lake Whatcom, Washington (U.S.A.) (Matthews et al., 2002). Al-Haidarey et al. (2016) used three parameters: TP concentration, Chl-a concentration, and SD, along with numerous physiochemical parameters: temperature, pH, DO concentration, electrical conductivity, HCO_3^- concentration, TN concentration, total hardness, Cl^- concentration, SO_4^{2-} concentration, total organic carbon concentration, Na^+ concentration, K^+ concentration, Fe^+ concentration, Mn^+ concentration, to determine Carlson's Trophic State Index (CTSI) of Bahr Al-Najaf reservoir, Iraq (Al-Haidarey et al., 2016). In 2017, Elshemy and Meon (2017) directly compared CTSI with LAWA trophic index and two other indices of water quality (NSF WQI and CCME WQI) in assessing the quality of water and trophic state in Aswan High Dam reservoir in Egypt (Elshemy and Meon, 2017).

The quality of water in reservoir usually accounts for its trophic state or degree of eutrophication. Carlson's Trophic State Index (CTSI), adopted by TEPA for reservoirs, is the most commonly used metric of water quality. Various investigation have used Carlson's Index to evaluate trophic states of water (Alemayehu and Hackett, 2016; Elmaci et al., 2009).

Carlson's Trophic State Index, which is a prevailing standard for representing a lake's trophic state or overall health, divides eutrophication into three levels. Water is in the oligotrophic state (excellent quality) if its CTSI is below 40, in mesotrophic state (good quality) if its CTSI is between 40 and 50, and of poor quality (algae breeding, possible algae poisoning and an odor) if its CTSI exceeds 50 (Carlson, 1977).

Three independent variables, chlorophyll-a concentration, total phosphorus concentration and Secchi depth, can be used to calculate Carlson's Index. The classical CTSI has the following formula (Carlson and Simpson, 1996).

$$\mathrm{CTSI} = \frac{\mathrm{TSI(SD)} + \mathrm{TSI(Chl\text{-}a)} + \mathrm{TSI(TP)}}{3} \quad (1)$$

$$\mathrm{TSI(SD)} = 60 - 14.41 \ln[\mathrm{SD}] \quad (2)$$

$$\mathrm{TSI(Chl\text{-}}a) = 30.6 + 9.8 \ln[\mathrm{Chl\text{-}a}] \quad (3)$$

$$\mathrm{TSI(TP)} = 4.15 + 14.42 \ln[\mathrm{TP}] \quad (4)$$

where TSI is the trophic state index of each water quality variable and [SD] is the Secchi depth in meters; [Chl-a] is the concentration of chlorophyll-a in μg/L and [TP] is the concentration of total phosphorus in μg/L.

Chlorophyll is the green molecule in plant cells that carries out the bulk of the energy fixation in the process of photosynthesis. Chlorophyll-a, one of the family of chlorophyll molecules, can be found in all plant cells and its concentration is therefore targeted by chlorophyll analysis. It is presented in all types of algae inside water bodies and is a photosynthetic pigment that indicates the productive processes and development of algae in a lake (Sellami et al., 2012). Chlorophyll-a concentration is a useful measure of water eutrophication.

Phosphorus is probably the most commonly studied plant nutrient in freshwater ecosystems because it frequently limits the growth and biomass of algae in lakes and reservoirs (Wetzel and G., 1975). Total phosphorus concentration in natural water is the sum of three phosphorus concentrations: soluble reactive phosphorus concentration,

soluble unreactive or soluble organic phosphorus concentration and particulate phosphorus concentration. Owing to its importance in limiting the development of algae and its necessity in the numerous empirical models, the total phosphorus concentration is the main variable to be measured in considering the threat of eutrophication in natural water.

A Secchi disk is a weighted disk with alternating black and white quadrants. The disk, used by the TEPA to measure the transparency of water in lakes or reservoirs, has a diameter of 30 cm. Brown and Simpson (2001) described the method of measuring Secchi depth as the following steps (Brown and Simpson, 2001). The disk is lowered into the water by a measured cord or rope until it cannot be seen. The depth (*i.e.*, the length of the rope from the disk to the water surface) is recorded. Then the disk is raised until it can be seen again. The average between the depth of disappearance and the depth of appearance, measured in meters, is called the Secchi depth. The Secchi depth is strongly correlated with the algae biomass in a water column in lakes in which sediment, silt, or other material does not affect clarity.

Among the three aforementioned metrics, chlorophyll concentration is the most accurate predictor of biomass (Carlson and Simpson, 1996). Phosphorus concentration may yield a more accurate estimate of a water body's summer trophic status than chlorophyll if the measurements are made during the winter. Secchi depth is probably the least accurate measure as it depends strongly on weather or natural conditions, but it is the most affordable and easiest to measure (Al-Haidarey et al., 2016). The TSI that is based on each of the above variables is not reliable when water turbidity is high (Chang and Liu, 2015).

Taiwan Environmental Protection Administration took water samples from reservoirs in Taiwan and analyzed them using the spectrophotometer method/vitamin C method (test method: NIEA W427.53B) to determine total phosphorus concentration and the acetone extraction method (test method: NIEA E507.03B) to determine chlorophyll-a concentration (Taiwan Environmental Protection Administration, 2017a). Before the spectrophotometer can be used to measure both of these parameters, chlorophyll-a must be extracted using 90% acetone through filter paper. Total phosphorus undergoes complex chemical reactions with several compounds, such as sulfuric acid, persulfate, ammonium molybdate, antimony potassium tartrate, and ascorbic acid thoroughly. To obtain the above parameters, water samples taken from reservoirs require sophisticated experiments that are carried out in the chemical laboratory.

Owing to the challenges associated with measuring concentrations of chlorophyll-a and total phosphorus, numerous other water quality indexes are utilized, including turbidity, electrical conductivity, total dissolved solid (TDS) concentration, biological oxygen demand concentration or biochemical oxygen demand (BOD) concentration, chemical oxygen demand (COD) concentration, and all of which are the most commonly used criteria for evaluating water quality (Chen et al., 2003; Najah et al., 2013; Thair et al., 2014; Seshan et al., 2014; Verma and Singh, 2013). Notably, Chang and Liu (2015) indicated that a trophic status determined using TDS, BOD, and COD is unreliable because of unstable turbidity, and proposed a fuzzy back-propagation neural network model to classify degree of eutrophication (Chang and Liu, 2015).

Not only are reliable models of evaluating water quality needed, but also demand for a more efficient approach is increasing. In recent years, various AI based methods have been used to solve water quality problems, and AI has great potential in the field (Chau, 2006). The essence of AI is the use of machine learning based on historical data to obtain functional relationships between inputs and outputs. Artificial neural networks, decision trees (DTs), linear regression, and the support vector machine are at the core of the most commonly used data mining algorithms for this purpose in related works. Some of them are utilized individually for prediction, as in ANNs based applications (Hamed et al., 2004; Maier and Dandy, 2000; Wu et al., 2014), and SVM based applications (Guo et al., 2014; Tarmizi et al., 2014; Zhao and Wang, 2010).

For comprehensive analysis, many researchers have combined the above techniques to construct an analytic system including comparison of predictive performances of multiple AI techniques. In the literature, ANNs are often utilized in water modeling and compared with other AI based techniques. In 2012, Couto et al. described the training, validation and application of ANNs and DTs to forecast the quality of water in the Odivelas reservoir in Portugal (Couto et al., 2012). Abyaneh (2014) and Chen et al. (2015) used ANNs and LR with the similar purpose (Chen and Liu, 2015; Zare Abyaneh, 2014). In 2014, Liu et al. indicated that SVM models are powerful alternatives to ANNs in predicting water quality in nonpoint source polluted rivers (Liu and Lu, 2014). Ji et al. (2017) developed an AI system with multiple models based on SVM, ANNs, LR to prove the superiority of SVM in predicting DO concentration in Wen-Rui Tang River (Ji et al., 2017). Therefore, this investigation uses the four AI techniques based on ANNs, SVM, LR, and Classification and Regression Trees to develop single (baseline) models and ensemble models using four software packages (RapidMiner Studio, Microsoft Azure Machine Learning Studio, WEKA, and IBM SPSS Modeler).

Least squares SVR (LSSVR), an alternative to SVM for regression, has also been used to improve predictions of water quality. Tan et al. (2012) and Kisi et al. (2016) compared the performance of LSSVR with that of other AI based techniques (Kisi and Parmar, 2016; Tan et al., 2012). They found that LSSVR is a more effective predictive tool than M5Tree, multi-layer BP and the RBF network in forecasting water quality.

Moreover, many researchers have used metaheuristic algorithms, such as genetic algorithms (GA), particle swarm optimization (PSO), the shuffled frog leaping algorithm (SFLA), ant colony optimization (ACO), and the artificial bee colony (ABC) algorithm, to enhance predictive AI models with good results in the related fields. In 2006, Chau (2006) and Kuo et al. (2006) integrated GA with ANNs for modeling water quality (Chau, 2006; Kuo et al., 2006). Zhang et al. (2011) used neural networks model, trained by PSO, to forecast storm water runoff discharge (Zhang and Lai, 2011). Chen et al. (2015) developed a hybrid neural network model, based on ACO and ABC, to forecast downstream river flow (Chen et al., 2015). Mahmoudi et al. (2016) optimized the SVR model using SFLA to predict the values of water quality parameters and proved that it is more efficient and powerful than the genetic programming model (Mahmoudi et al., 2016).

Recent works have proposed a new biologically inspired metaheuristic algorithm, known as the firefly algorithm (FA) which have been developed by Yang (2008) (Yang, 2008), as an optimization algorithm with various applications (Yang, 2010, 2011, 2013). The FA has been proven to be better than other optimization algorithms, such as PSO and GA (Pal et al., 2012). Banati and Bajaj (2011) found that FA was faster and better than GA, ACO, PSO and ABC, in feature selection (Banati and Bajaj, 2011). Chou et al. (2016) demonstrated the superiority of LSSVR models that are based on smart FA, in terms of in computational efficiency and solution quality, for solving civil engineering problems (Chou et al., 2016b).

AI techniques are still difficult to implement in the real world. An expert system with a window based or browser based interface is necessary to reduce the need for users to practice the application of such a technique. Zapata et al. (2013) presented an interface to assist instructors, students, and professionals in sharing knowledge about engineering problems and projects (Zapata et al., 2013). Lin (2013) developed a construction network based interface management system for information sharing and tracking efficiency (Lin, 2013). In 2016, Chou et al. constructed a nature-inspired metaheuristic regression system that includes a smart FA based least squares SVR (SFA-LSSVR) model with manual operations through a user-friendly interface (Chou et al., 2016a).

Although the efficiency of FA that is integrated with the LSSVR model has been proven in several fields, such as electrical load (Kavousi-Fard et al., 2014), geosynthetic-reinforced soil structures

(Chou et al., 2015), steel pitting risk, and corrosion rate (Chou et al., 2017), this method has not been utilized to predict the quality of water. Therefore, this work combines a least-square SVR model with an enhanced FA that is optimized by supplement metaheuristic components (*i.e.*, chaotic maps, adaptive inertia weight, and Lévy flight) to construct a water quality prediction model, which is applied using a user-friendly interface. The performance of this hybrid metaheuristic regression model is compared with those of other AI models, implemented in four software packages, with the ultimate goal of comprehensively assessing the quality of water in 20 reservoirs.

3. Methodology

In this study, baseline models, ensemble models, and a hybrid metaheuristic regression model were proposed to predict the quality of water using five data mining software packages, which are RapidMiner Studio, Azure Machine Learning Studio, IBM SPSS Modeler, WEKA and MATLAB. Baseline models are used to identify the best of a diverse range of single models that are generated in the first four packages. With the same purpose, ensemble models were analyzed in the software; these models were based on the models that individually yielded the best results. Lastly, MATLAB was used to establish a hybrid model, which was compared with baseline and ensemble models. The best performing of the three approaches was identified.

3.1. Baseline models

3.1.1. Artificial neural networks

Artificial neural networks, which are commonly referred to as neural networks, mimic the neuro system for computational purposes. The basic structure of a neural network consists of three main layers of interconnecting artificial neurons; the layers are an input layer, a hidden layer and an output layer. A neural network uses inter-connected neurons, which send the signals within the network, as processing elements (Singh et al., 2007). Fig. 1 depicts the inter-connection between neurons of layers.

To train an ANN, the virtual network inputs data from the data-set to the neurons of the first layer. These neurons, simply called nodes, have two main functions, which are to adjust the node weight by its adaptive learning from the input data and convert it into an impulse, and to transmit that impulse to other nodes. When the whole system performs the training process, each neuron receives the input from the neuron in the former layer and its output is sent as an input to the neuron in the next layer in the network. Accuracy of outcomes generated by network in the training process is improved as the network learns. This procedure is repeated until a stopping criterion (*i.e.*, 100 epochs or the error threshold $\leq 10^{-5}$ in this study) is met.

3.1.2. Support Vector Machine for classification and regression

The Support Vector Machine (SVM) is a supervised learning technique that was invented by Vapnik (1995) (Vapnik, 1995). The SVMs can be used for both classification and regression problems (Chou et al., 2016b). The Support Vector Machine for Classification is a technique that determines an optimal hyper plane to separate two classes from the training set.

The Support Vector Regression (SVR) is a variation of the SVM that is used to construct an input-output model for solving nonlinear regression problems. In SVR, the input is first mapped onto n-dimensional feature space by a fixed mapping procedure. A linear model is then constructed in this feature space which can be expressed by Eq. (5)

$$f(x,\omega) = \sum_{j=1}^{n} w_j g_i(x) + b \tag{5}$$

where $g_j(x)$ is a set of non-linear transformation and b is a bias term. A loss function $L(x,\phi)$ is then used for measuring estimation quality as follows:

$$L_\varepsilon = [y, f(x,\omega)] = \begin{cases} 0 & if\ |y - f(x,\omega)| \leq \varepsilon \\ |y - f(x,\omega)| & otherwise \end{cases} \tag{6}$$

The most novel feature of SVR is its use of ε-insensitive loss to compute a linear regression function for a new higher-dimensional feature space while simultaneously reducing the complexity of model by minimizing $\|\omega\|^2$.

During training, SVR kernel functions (linear, radial basic, polynomial, or sigmoid) are used to identify support vectors along the function surface. Since the RBF kernel maps samples non-linearly into a higher dimensional space and presents fewer numerical difficulties, RBF is selected for SVR kernel function (Chou et al., 2016b; Chou et al., 2016c).

3.1.3. Classification and Regression Trees

Classification and Regression Tree analysis commonly refers to both of classification and regression trees according to the type of dependent variable, which may be categorical or numeric (Breiman et al., 1984). A decision tree learning process involves the construction of a decision tree from class-labeled training data. A decision tree represents the test on each attribute by an internal node. The outcome of the test is represented by a branch and each leaf node has a class label or prediction (Kumar, 2013). The CART model performs classification through the recursive splitting process. The tree is pruned to optimize the prediction accuracy by minimizing the number of branches according to the Gini index (Chou et al., 2016c). Then the analysis results are evaluated *via* performance metrics.

3.1.4. Linear regression

Linear regression is a regression model that was developed to

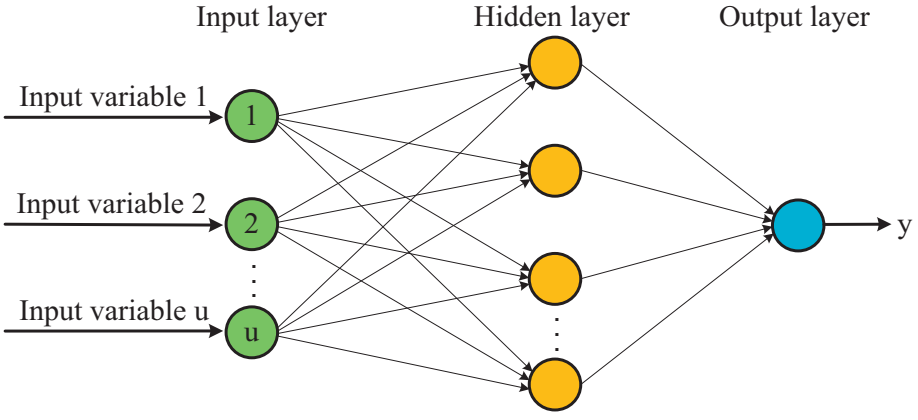


Fig. 1. Layered structure of ANNs model.

establish the relationship between independent and dependent variables in a prediction problem (Chou et al., 2016c). In this study, multivariate linear regression, which is one of several versions of linear models, is used. The water quality dataset involves a numerical response variable, and explanatory variables that need to be evaluated by multivariate linear regression for the purpose of making a prediction. In this investigation, ordinary least squares method is used to fit regression line in the software packages.

3.1.5. Software

This work aims to assess both predictive accuracy and applicability of models for the particular demand of practitioners. Therefore, the software packages that provide the user-friendly interface and powerful predictive applications are necessary. To obtain accurate predictions from the baseline models, the commonly used data-mining packages RapidMiner Studio, Azure Machine Learning Studio, IBM SPSS Modeler and WEKA were used for the ease of machine learning implementation.

3.2. Ensemble models

Ensemble systems use a strategy of creating many single AI based models and metaheuristically combine their outputs such that the combination outperforms the single models. Particularly, a combination of single models can compensate for errors made by the single models in different parts of the input space. Their performances then proved the dominance over those in baseline cases (Chou et al.; Chou et al., 2016c). Therefore, the following ensemble methods are used herein to obtain improved predicted outputs.

3.2.1. Voting

Voting is the simplest method of combining multiple AI techniques (Kittler et al., 1998). In prediction problems, the outputs of individual models are pooled, and thus, the class of the largest number of votes is selected for the further analysis. Generally, the numerical output can be determined by different combinations of probability estimates. In this study, voting method was utilized to generate five ensembles of three or four different individual techniques. The three-technique ensembles included ANNs + CART + SVR, ANNs + CART + LR, CART + SVR + LR, and ANNs + SVR + LR. The one four-technique ensemble was: ANNs + CART + SVR + LR.

3.2.2. Bagging

In the bagging method, bootstrapping is used to train several models independently and with different training set (Breiman, 1996). Bootstrapping builds *n* replicate training datasets to construct *n* independent models by randomly re-sampling and replacing the original training dataset. Subsequently, each training sample may appear repeatedly or not at all in any particular replicated training dataset of *n*. Then, an appropriate combination method, such as average of probabilities, is implemented to aggregate *n* single models. This study proposes four homogenous ensembles, generated using individual learning techniques, they are an ANNs ensemble, an SVR ensemble, a CART ensemble, and an LR ensemble. In the study, the number of independent models is selected to be 10 to reduce possible bias in the re-sampling and replacing the original training dataset.

3.2.3. Stacking

Stacking or stacked generalization hierarchically constructs multilevel classifiers in which first level consists of several single classifiers. The outputs of the first-level classifiers are then used to train the second level classifier, which is called a 'stacked' classifier (Wolpert, 1992). Therefore, the final output depends on the output of the stacked classifier. In the stacking method, the stacked classifier, unlike the 'static' classifiers in the above ensemble methods, is a 'trainable' combiner, meaning that it identifies classifier errors for a particular learning dataset and corrects them.

The stacking based ensemble system in this study is based on a two-level scheme with four experimental combinations of three heterogeneous techniques in the first level (ANNs + CART + SVR, ANNs + CART + LR, CART + SVR + LR, and ANNs + SVR + LR) and an experimental combination of all four techniques in the first level (ANNs + CART + SVR + LR). Since SVR is better than many other supervised learning techniques in solving many pattern recognition problems as a baseline classifier (Wolpert, 1992), all five ensembles used SVR in the second level of the streams of stacking models.

3.2.4. Tiering

Tiering involves analyzing data that is divided in two tiers. The first tier divides the data into *k* classes according to the standard value (*T* value), and the second tier establishes the *k* regression models to make predictions. The general equation for dividing the data according to the *T* value (Chou and Tsai, 2012) is $T = \frac{Y_{\max} + Y_{\min}}{k}$. Due to the context of distribution of the CTSI values in the data set, two classes ($k = 2$) was the optimal number of classes for predicting the values of the water quality index herein. Data samples were divided into two classes, which are high class and low class, using the rule that is expressed as Eq. (7).

$$T = \frac{Y_{\max} + Y_{\min}}{2} \tag{7}$$

where $Y_{act} \le T$ belongs to Class 1, and $Y_{act} > T$ belongs to Class 2. Once the training dataset is divided into two classes, the original output, Y_{act}, is replaced by a discrete class label corresponding to its class. The new training dataset is then used to construct a particular classifier. This study proposed SVMs as the particular classifier, and the output variable of each data sample is a discrete class label. When building two regression models of the second tier, however, the Class *j* subset is used to train the jth regression model ($j = 1$, *or 2)*, and the output variable of each data sample in the Class *j* subset is the actual value (CTSI).

After the tiering models were trained and constructed, the test data were used to evaluate their performance. Let T_i be the *i*th data sample in the testing dataset where $T_i = (T_1, T_2, \ldots, T_n)$ contains n input variables. In the first level of the tiering model, T_i is input into the SVM classification model in which it is classified as either low or high class. Consequently, the data sample T_i is input to the trained regression model that corresponds to its class label. Finally, the output variable of sample T_i (*i.e.*, CTSI) is predicted by the corresponding 'high'-regression models or 'low'-regression models (such as SVR, ANNs, CART, and LR). The average value of performance measure is formulated according to the number of classes: ((number of data samples in the first class/total number of data samples) × value of measure in first class) + ((number of data samples in the second class/total number of data samples) × value of measure in second class).

3.2.5. Implemented software

In this study, the software that performed best in the baseline scenario is the IBM SPSS Modeler, and this was used to construct the ensemble models. Although this software offers an easily used platform for developing basic individual AI models, advanced techniques, such as ensemble models proposed in this study, are quite customized based on the novelty of the experienced users.

3.3. Hybrid model

3.3.1. Least squares support vector machine for regression

The regression model of SVMs, support vector machine based regression (SVR), is widely used to construct an input-output model in the solution of nonlinear regression problems. The ε-insensitive loss function, which is the unique feature of SVR, is one of two important parameters (C, ε) that must be set by users. The hyper-parameter ε may be eliminated by replacing the original ε-insensitive loss function with a least-square loss function (Suykens and Vandewalle, 1999). The least-

square loss function was herein modified to a support vector machine in the formulation of LSSVR for estimating nonlinear function, as proposed by Suykens et al. (2002) (Suykens et al., 2002).

LSSVR uses equality constraints instead of inequality constraints, which are used by SVR, and it applies a quadratic error criterion. SVR involves solving a quadratic programming problem with linear inequality constraints whereas LSSVR involves solving a set of linear equations, facilitating implementation of LSSVR (Kuh, 2004). LSSVR is a highly developed machine learning technique with many advanced features that enable good generalization capacity and fast computation.

In the training process, the LSSVR model utilizes a least squares loss function to obtain a linear set of equations in the dual space. Compared to SVR, LSSVR is easier to implement and can be implemented adaptively. Whereas SVR solutions are typically found using batch algorithms, LSSVR solutions can be found using on-line adaptive algorithms (Kuh, 2004), simplifying its use in adaptive signal processing and communication. Accordingly, solutions are derived by using iterative methods, such as conjugate gradient, to solve a set of linear equations efficiently.

Although LSSVR effectively solves prediction problems, a major drawback is that the accuracy of an LSSVR model depends on the regularization and kernel parameters. In highly nonlinear spaces, using a radial basis function (RBF) kernel, in the LSSVR usually yields more promising results than other kernels (Wang and Hu, 2005). To improve predictive accuracy, automatic optimization in LSSVR should involve the regularization parameter (C) and the sigma of the RBF kernel (σ). Therefore, this study proposed the following metaheuristic algorithm to optimize the LSSVR parameters.

3.3.2. Metaheuristic optimization algorithm

Developed by Yang (2008) (Yang, 2008), the Firefly Algorithm (FA) is a stochastic, nature-inspired metaheuristic algorithm that is effective in solving optimization problems (Fister Jr et al., 2015). The FA follows three idealized rules: (1) a firefly is attracted to other fireflies regardless of gender; (2) the attractiveness of a firefly decreases as distance increases and a firefly moves randomly if nothing else is brighter than itself; and (3) the brightness of a firefly is affected or determined by the landscape of the objective function.

Although the basic FA is very efficient, the solutions continuously change as it approaches the optima (Pal et al., 2012). The firefly algorithm involves a local search and sometimes cannot escape a local optima (Farahani et al., 2011). Hence, an important aspect of swarm intelligence and modern meta-heuristics is randomization, which enables an algorithm to jump out of any local optimum during a global search. Fine-tuning the randomness and balancing local search against global search are essential to the favorable performance of a metaheuristic algorithm.

The conventional FA utilizes three metaheuristic components in performing a heuristic search that proceeds toward global optimality. These components are the initial population, the attractiveness parameter, and movement. In this study, these three terms are adjusted using supplementary components, which are chaotic maps, adaptive inertia weight (AIW), and Lévy flight, to improve the ability of the FA to find the global optimum (Chou and Pham, 2015). The aim of the modification is to maintain balance between exploration and exploitation and to increase the probability of escaping from the basins of attraction of local optima. The proposed version of FA, MetaFA, is improved by adding the following smart components.

3.3.2.1. Chaotic maps. Recent theoretical advances in, and applications of, nonlinear dynamics, and especially chaos theory, have attracted attention in many fields. One application of chaos theory is in optimization algorithms (Yang et al., 2007). For example, different chaotic maps have been proposed as alternatives to pseudorandom sequences. Comparisons show that the Gauss/Mouse map (He et al., 2001) is the best chaotic map for tuning the attractiveness parameter in the original FA.

The initial population of an optimization algorithm, such as FA, must be sufficiently diverse to prevent the search from trapping in the local optima. Logistic mapping (Li et al., 2011), which is the simplest chaotic mapping operator, can provide more diversity than randomly selected initial populations and can reduce the probability of the premature occurrence.

3.3.2.2. Adaptive inertia weight. The inertia weight affects both the convergence to the optimal solution and the execution time of the simulation. A high inertia weight favors global exploration in the early stages of the search process whereas a low inertia weight favors local exploration in the late stages. The inertia weight controls the local and global exploration capacities of the swarm algorithm.

The dynamic exploration is addressed by using a monotonically decreasing function to change the inertia weight and the attractiveness coefficient. In this study, parameter was tuned with an adaptive inertia weight to reduce the attractive distances and maintain it in a reasonable range.

3.3.2.3. Lévy flight. Random walk theory has a central role in modern meta-heuristic algorithms and stochastic optimization. Lévy flights are random walks with a power-law distribution of step-length. To generate the steps, the Mantegna's algorithm is used herein to provide a symmetric Lévy stable distribution.

MetaFA is more effective and efficient than the standard FA, because the random steps are replaced by chaotic maps and Lévy flights. In MetaFA, the logistic chaotic map is used to generate a highly diverse initial population. Next, the attractiveness parameter is tuned using the Gauss/mouse chaotic map. Then, the AIW is used to tune the randomization parameter, which controls local exploitation and global exploration. Finally, Lévy flight is used to generate new solutions around the optimal solution accelerating the exploitation.

3.3.3. Constructing MetaFA based LSSVR model

The proposed LSSVR model employs a least-square loss function to reduce the computational burden. The RBF kernel function is used because it reportedly outperforms other kernel functions for LSSVR in highly non-linear spaces (Wang and Hu, 2005). To increase the accuracy of forecasting, two parameters in the LSSVR model must be specified for obtaining the optimal results. These parameters are as follows; (1) regularization parameter (C) for determining the trade-off between minimizing the training error and minimizing model complexity; (2) the sigma parameter of the RBF kernel function (σ), which defines the non-linear mapping from the input space to the high-dimensional feature space.

The optimization of the LSSVR parameters is performed using the meta-heuristic algorithm (FA), which involves three supplementary components (*i.e.*, chaotic maps, adaptive inertia weight, and Lévy flights). These components are used for adjusting the conventional meta-heuristic algorithm, which are initial population, attractiveness coefficient, and movement. The modifications of FA to optimize LSSVR hyper-parameters yields an advanced hybrid model, MetaFA-LSSVR, which could efficiently solve the above problems.

Historical data were classified into two datasets, which were the learning set for building models, and a testing set for evaluating the performance of the trained LSSVR model after it was optimized with unseen data. The optimization was automatically conducted by utilizing MetaFA to optimize the LSSVR parameters simultaneously. LSSVR involves mainly learning and curve fitting whereas MetaFA optimizes parameters C and σ to minimize the prediction error.

3.3.4. Evaluation of optimization algorithm

Metaheuristic algorithms involving optimization approach should be evaluated using actual cases with known solutions. Benchmark functions, which define the artificial landscapes in applied

Table 2
Benchmark functions for evaluating optimization algorithm.

No.	Name	Function	Dimension	Optimum	Mean	Standard deviation	Average time per run (s)
1	Stepint	$f_1(x) = 25 + \sum_{i=1}^{D} (\lfloor x_i \rfloor)$	10	0	1.55E+01	5.09E−01	2.43E+02
		x = [−5.12;5.12]	30	0	−3.33E+00	7.11E−01	3.78E+02
		Minimum $f_1(0, \ldots,0) = 0$	50	0	−2.08E+01	1.60E+00	4.91E+02
2	Powell Sum	$f_2(x) = \sum_{i=1}^{D} \lvert x_i \rvert^{i+1}$	10	0	2.54E−08	2.07E−08	1.37E+03
		x = [−1;1]	30	0	1.06E−04	2.32E−04	1.11E+03
		Minimum $f_2(0, \ldots,0) = 0$	50	0	6.96E−05	1.92E−04	1.98E+03
3	Quintic	$f_3(x) = \sum_{i=1}^{D} \lvert x_i^5 - 3x_i^4 + 4x_i^3 + 2x_i^2 - 10x_i - 4 \rvert$	10	0	1.14E−04	2.17E−05	8.64E+02
		x = [−10;10]	30	0	1.05E−03	2.11E−04	1.17E+03
		Minimum $f_3(0, \ldots,0) = 0$	50	0	4.60E−01	8.55E−01	1.30E+03
4	Schumer Steiglitz	$f_4(x) = \sum_{i=1}^{D} x_i^4$	10	0	6.62E−23	5.04E−23	8.47E+02
		x = [−100;100]	30	0	6.18E−20	2.59E−20	1.07E+03
		Minimum $f_4(0, \ldots,0) = 0$	50	0	2.01E−11	1.10E−10	1.36E+03
5	Styblinski-Tang	$f_5(x) = \frac{1}{2} \sum_{i=1}^{D} (x_i^4 - 16x_i^2 + 5x_i)$	10	−3.92E+02	−9.18E+01	3.64E+00	8.50E+02
		x = [−5;5]	30	−1.17E+03	−2.50E+02	1.32E+01	1.17E+03
		Minimum $f_5(-2.903534, \ldots, -2.903534) = -39.16599D$	50	−1.96E+03	−3.70E+02	1.71E+01	1.28E+03

mathematics, are often utilized to evaluate metaheuristic algorithms. A range of recognized benchmark functions have been designed to test global optimization algorithms. Table 2 presents several classic benchmark functions that are used herein to evaluate the performance of the optimization method in this study.

3.3.5. Implemented Software

The hybrid MetaFA based LSSVR model was developed in MATLAB, which is a mathematical software that was developed by MathWorks. The software allows the user to work with vectors and matrices directly and provides extensive functionality for numerical calculations. The main part of the software is the interpreter, which interprets commands in the MATLAB programming language. These commands, called M-Code, are typed by the user. MATLAB has proved its usefulness for rapid analysis and data processing.

3.4. Cross-validation method

The stratified cross-validation procedure is utilized to compare prediction accuracies of two or more models. A k-fold cross-validation method can minimize bias that is associated with random samples of training and testing data. This method initially splits the samples into k mutually exclusive subsets. Each time, k-1 subsets are used as training samples, and the remaining subsets are used as test samples. This model validation process is carried out k times to reduce errors during random sampling. Kohavi (1995) proved that ten-fold validation is the optimal choice to obtain analytical validity, computational efficiency, and optimal deviation (Kohavi, 1995). Therefore, each of ten folds will take turns as the testing fold while the other nine folds will be combined as the training dataset to create a prediction model. The performance of each proposed model was evaluated in terms of average prediction error in ten fold validation.

3.5. Performance evaluation

To assess the predictive accuracy of the proposed models, five metrics are used; they are R, MAPE, RMSE, MAE and SI.

3.5.1. Linear correlation coefficient (R)

The quantity R, called the linear correlation coefficient, measures the strength and direction of a linear relationship between two variables. The mathematical formula for R is:

$$R = \frac{n\sum y.\, y' - (\sum y)(\sum y')}{\sqrt{n(\sum y^2) - (\sum y)^2}\sqrt{n(\sum y'^2) - (\sum y')^2}} \tag{8}$$

where y' = predicted value; y = actual value; and n = number of samples.

3.5.2. Mean Absolute Percentage Error (MAPE)

The MAPE is effective for identifying differences between models because it is unaffected by actual and predicted values or their units. The MAPE can be calculated using Eq. (9).

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y - y'}{y}\right| \tag{9}$$

where y' is the predicted value; y is the actual value; and n is the number of data samples.

3.5.3. Root Mean Squared Error (RMSE)

The RMSE is the square root of the sum of square errors between predictions and actual values. The RMSE is therefore the mean vertical distance of a data point from a fitted line. It efficiently reveals undesirably large differences and is given by Eq. (10).

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y' - y)^2} \tag{10}$$

where y' is the predicted value; y is the actual value; and n is the number of data samples.

3.5.4. Mean Absolute Error (MAE)

The MAE is a measure of closeness of forecasts to outcomes outcomes. It is the average magnitude of errors between predicted and actual values, regardless of their direction. Eq. (11) is the mathematical formula for the MAE.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y - y'| \tag{11}$$

where y' is the predicted value; y is the actual value; and n is the number of data samples.

3.5.5. Synthesis Index

The synthesis Index (SI) based on the four statistical measures, 1-R, MAPE, RMSE, and MAE, is derived as a comprehensive performance

Table 3
Water quality data description.

Field	Value
Dataset characteristics	Multivariate
Attribute characteristics	Real
Associated tasks	Regression
Number of instances	1635
Number of attributes	8
Number of dummy attributes	22
Missing values	1707
Time	1995–2016
Removed attributes	Storage Capacity, Total Organic Carbon (TOC)

Table 4
Descriptive elements of water quality dataset.

Variable	Unit	Parameter (abbreviation)	Direction	Min.	Max.	Mean
X1	˚C	Surface water temperature	Input	12.38	33.35	24.98
X2	mg/L	Biochemical Oxygen Demand (BOD)	Input	1.23	14.40	8.24
X3	mg/L	Suspended Solid (SS)	Input	0.70	143.48	6.03
X4	mg/L	Chemical Oxygen Demand (COD)	Input	0.50	108.78	8.01
X5	mg/L	Ammonia (NH_3)	Input	0.01	8.67	0.21
D6–D27	–	Dummy variables for season and location	Input	–	–	–
Y1	–	Carlson's Trophic State Index (CTSI)	Output	23.78	85.83	48.74

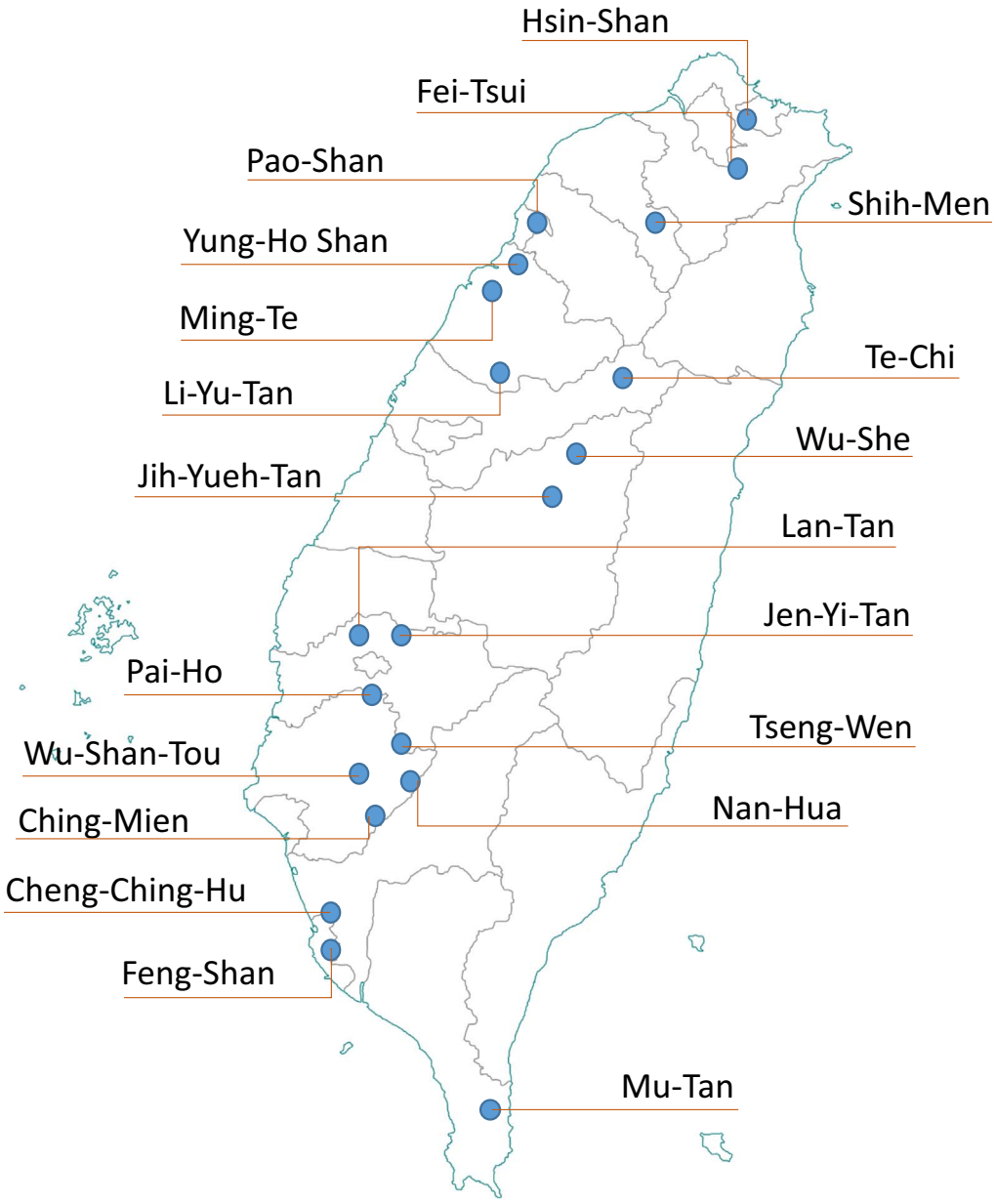


Fig. 2. Geographical locations of reservoirs.

Table 5
Default settings of model parameters.

Model	Parameters	Default value			
		Microsoft Azure Machine Learning Studio	RapidMiner Studio	IBM SPSS Modeler	WEKA
Artificial Neural Networks	Name	Neural Network Regression	Neural Net	Neural Net	Multilayer Perceptron
	Hidden layer specification	Fully-connected case	N/A	Simple	N/A
	Number of hidden layers	1	1	1	1
	Number of hidden nodes	100	$1+\frac{\text{atts}+\text{classes}}{2}$	20	$\frac{\text{atts}+\text{classes}}{2}$
	Number of learning iterations	100	500	200	500
	Momentum	0	0.2	0.9	0.2
	Type of normalize	[0,1]	[−1.1]	N/A	[−1.1]
	Shuffle examples	True	True	N/A	True
	Learning rate	0.005	0.3	Initial: 0.3 High: 0.1 Low: 0.01	0.3
Support Vector Machine	Name	Not available	SVM	SVM	SMOreg
	Kernel type	N/A	RBF	RBF	RBF
	Regularization parameter (C)	N/A	0	10	1
	ε (epsilon)	N/A	0.001	0.1	0.001
	RBF γ (gamma)	N/A	1.0	0.1	0.01
	Stopping criteria	N/A	N/A	1.0E−3	Tolerance: 0.001
Classification and Regression Trees	Name	Decision Forest Regression	W-REPTree	C&R Tree	REPTree
	Tree depth	32	Infinite	5	Infinite
	Number of random splits/leaf node	128	N/A	N/A	N/A
	Min number of samples/leaf node	1	2	Min records in parent branch: 2% Min records in child branch: 1%	2
	Create trainer mode	N/A	Single parameter	N/A	Single parameter
	Minimum variance for split (V)	N/A	1.0E−3	N/A	1.0E−3
	Number of folds for reduced error pruning (N)	N/A	3	N/A	3
	Seed for random data shuffling (S)	N/A	1	N/A	1
	Pruning (P)	N/A	True	True	True
Linear Regression	Name	Linear Regression	Linear Regression	Regression	Linear Regression
	Fitting regression line method	Ordinary Least Squares	Ordinary Least Squares	Ordinary Least Squares	Ordinary Least Squares
	L2 regularization weight	0.001	1.0E−8	N/A	1.0E−8
	Features selection	N/A	M5-prime	Only use complete records	M5-prime
	Use bias	True	True	N/A	N/A
	Eliminate collinear features	N/A	True	N/A	True
	Tolerance	N/A	0.05	1.0E−4	N/A

measure:

$$\mathrm{SI}=\frac{1}{m}\sum_{i=1}^{m}\left(\frac{P_i-P_{\min,i}}{P_{\max,i}-P_{\min,i}}\right) \tag{12}$$

where m is the number of performance measures, and P_i is the ith performance measure. The range of SI is from 0 to 1; an SI value close to 0 indicates that the predictive model is highly accurate.

4. Water quality data and parameters

4.1. Data

The dataset that is used herein was collected from measurements made at the stations of 20 reservoirs in Taiwan. Monitoring data (Table 3) recorded in a period of 22 years from the year 1995 to the year 2016 were preprocessed in this study. Fig. 2 illustrates the geographical locations of 20 reservoirs in Taiwan. Instead of acquiring complex variables, such as total phosphorus concentration, chlorophyll-a concentration, Secchi depth and pH, this investigation used easily collected water parameters to model the non-linear patterns between the variables and water quality index.

Specifically, the dataset includes seven attributes that can be used as inputs in the prediction of Carlson's Trophic State Index, which quantifies water quality. Among the attributes, since location and season affect the prediction results but are non-numeric feature, they were converted into dummy variables. Total Organic Carbon (TOC) and Reservoir Storage Capacity involved too many missing data points, and so were removed from the dataset along with a row containing a missing concentration of ammonia nitrogen (NH_3-N). The remaining dataset comprised 1635 data concerning 27 attributes, represented numerically, and this was used to generate multivariate machine learning models.

4.2. Parameters description

In the literature, the parameters, surface water temperature, BOD, COD, DO, ammonia nitrogen (NH_3-N) concentration, and suspended solids (SS), are the important factors in evaluating trophic state index and water quality. Liou et al. (2004) demonstrated that temperature, BOD, NH_3-N concentration, and SS are among the most important variables in determining water quality (Liou et al., 2004), as validated by the TEPA. Of the six aforementioned variables, BOD and COD are the two most popularly predicted as measures of water quality. In particular, Verma and Singh (2013) and Zare Abyaneh (2014) used temperature and concentration of SS, along with other indices, to predict BOD and COD. Tarmizi et al. (2014) developed an SVM model with five parameters, including temperature and the concentration of ammonia nitrogen (NH_3-NL), to forecast water quality output in terms of DO concentration (Tarmizi et al., 2014).

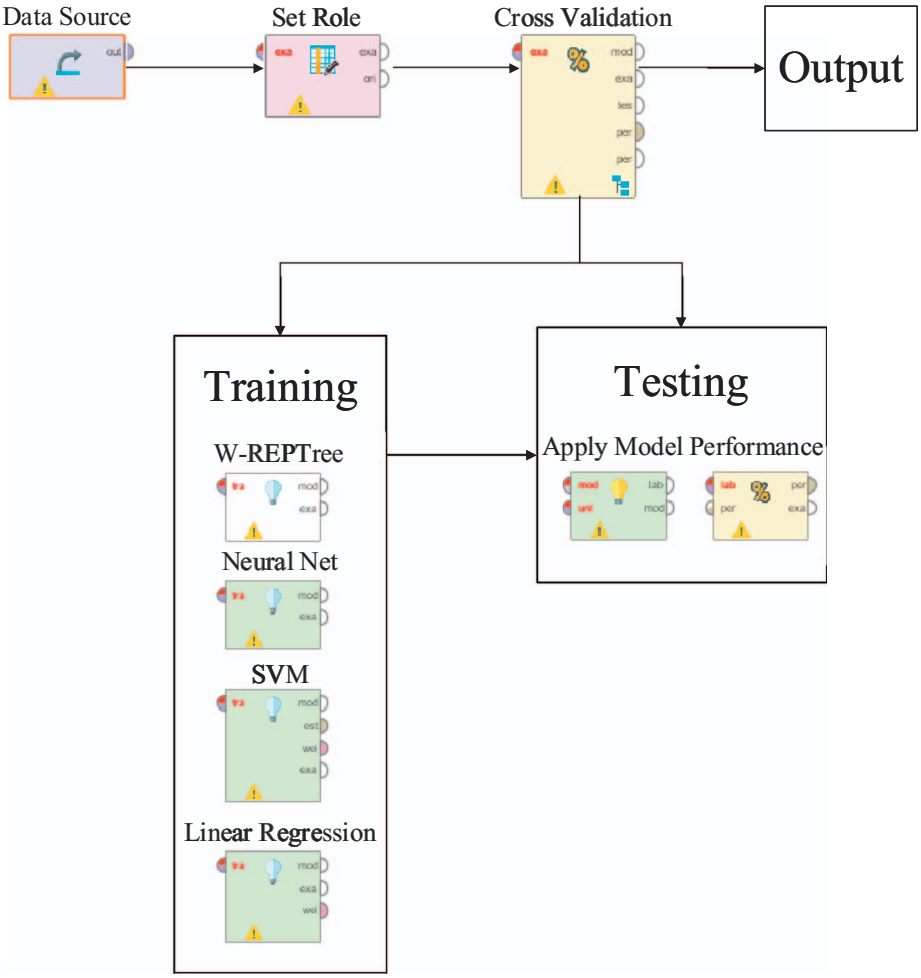


Fig. 3. Stream of baseline model in RapidMiner Studio.

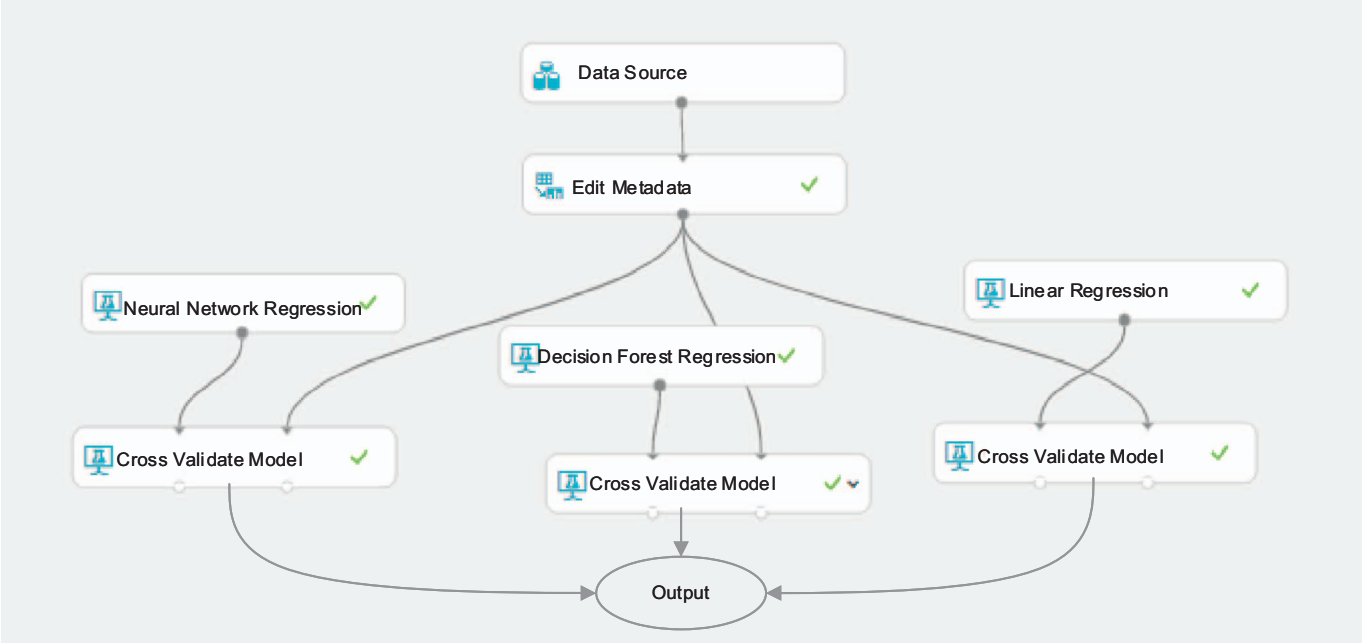


Fig. 4. Stream of baseline model in Microsoft Azure Machine Learning Studio.

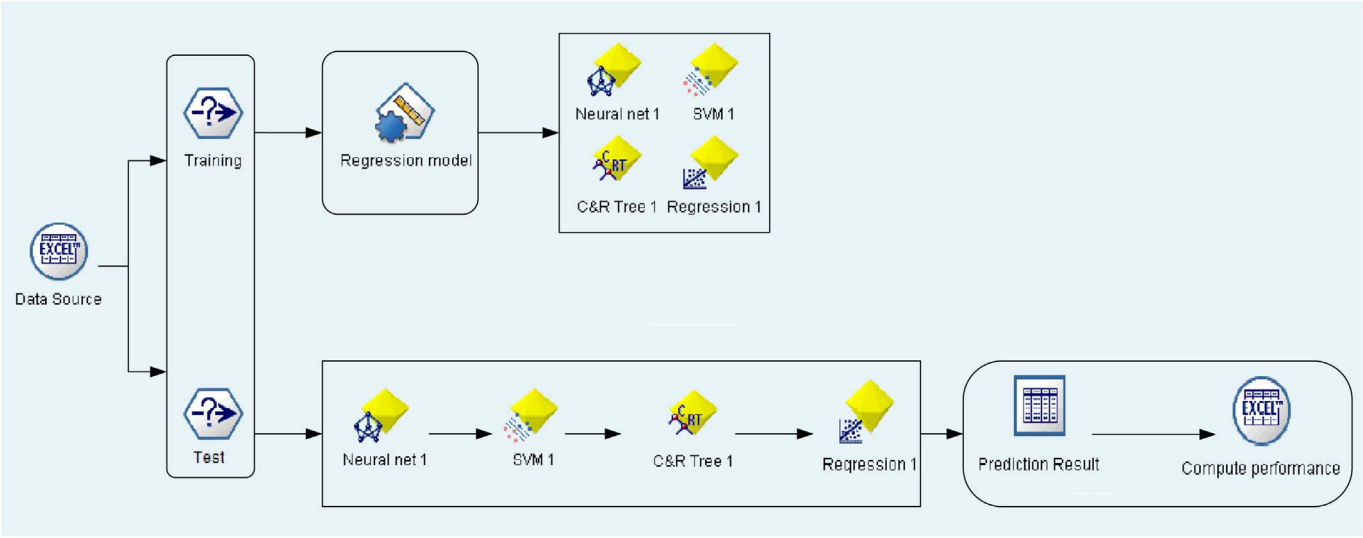


Fig. 5. Stream of baseline model in IBM SPSS Modeler.

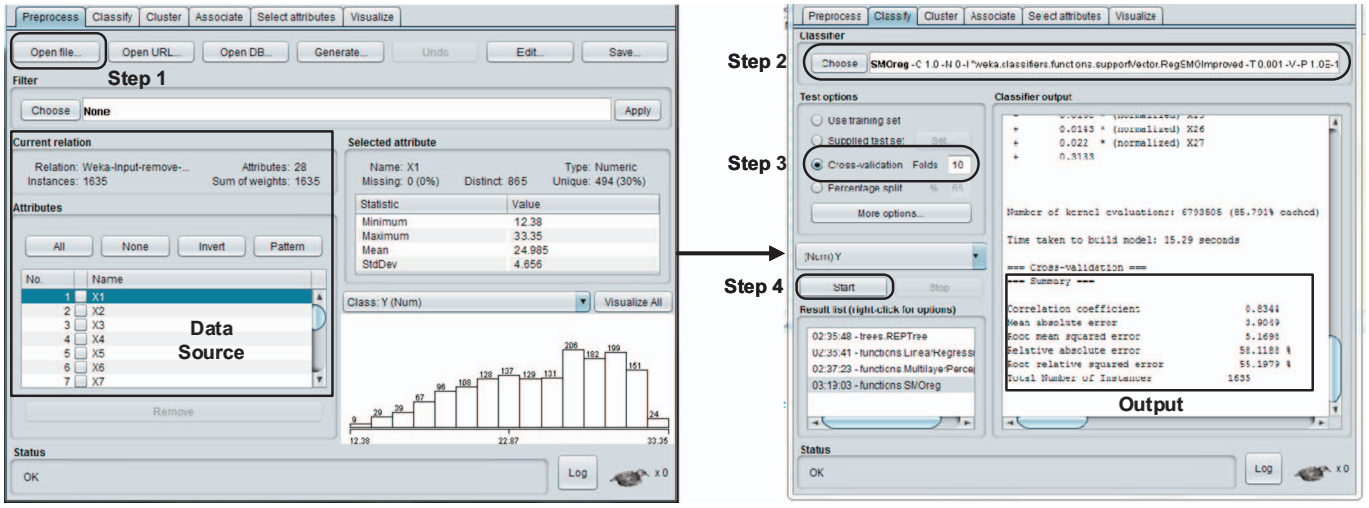


Fig. 6. Implementation of baseline models in WEKA.

Since the usefulness of these parameters in conventional water quality approaches have been proved, this study utilizes surface water temperature, BOD, SS, COD, NH_3 concentration, and SS in the predictive system. Location and season are also the important variables associated with the quality of water in reservoirs. The original dataset contains categorical values of season attribute and categorical location attributes that were then converted into numerical values using dummy attributes for subsequent regression modeling. Finally, to preprocess the input data, the five water parameters are associated with five corresponding input variables from X1 to X5, and 22 dummy attributes (*i.e.*, 3 for season variable and 19 for location variable) from D6 to D27 are set. The Carlson's Trophic State Index (CTSI), which is the output metric of water quality, is the target variable to be forecasted using these 27 input variables. Table 4 presents the descriptive elements of the water quality dataset.

5. Model development

5.1. Baseline models

Although all four software packages provide a user-friendly interface for the development of AI techniques, the baseline models in each software contain various parameter fields with different settings. For efficient operations and ease of use, the default settings of the main parameters were selected in each model in the four software packages. Table 5 presents the comprehensive settings for these individual models.

Microsoft Azure ML Studio, RapidMiner Studio and SPSS Modeler allow the user to adjust the structure of a baseline model manually by visualizing machine learning techniques as modules. The WEKA software executes analytic procedures, providing a graphical interface using which data are input and the process is begun by the push of a button. Figs. 3, 4, and 5 show the streams for implementing the baseline scenario in RapidMiner Studio, Microsoft Azure ML Studio and IBM SPSS Modeler, respectively. Fig. 6 presents the implementation procedures of baseline models in WEKA.

5.2. Ensemble models

As in the baseline scenario, user can construct ensemble analytic streams by arranging and customizing the graphical modules of AI techniques. The best software for implementing the baseline scenario was the IBM SPSS Modeler, which is therefore used to increase the predictive accuracy in the ensemble scenario. The following figures

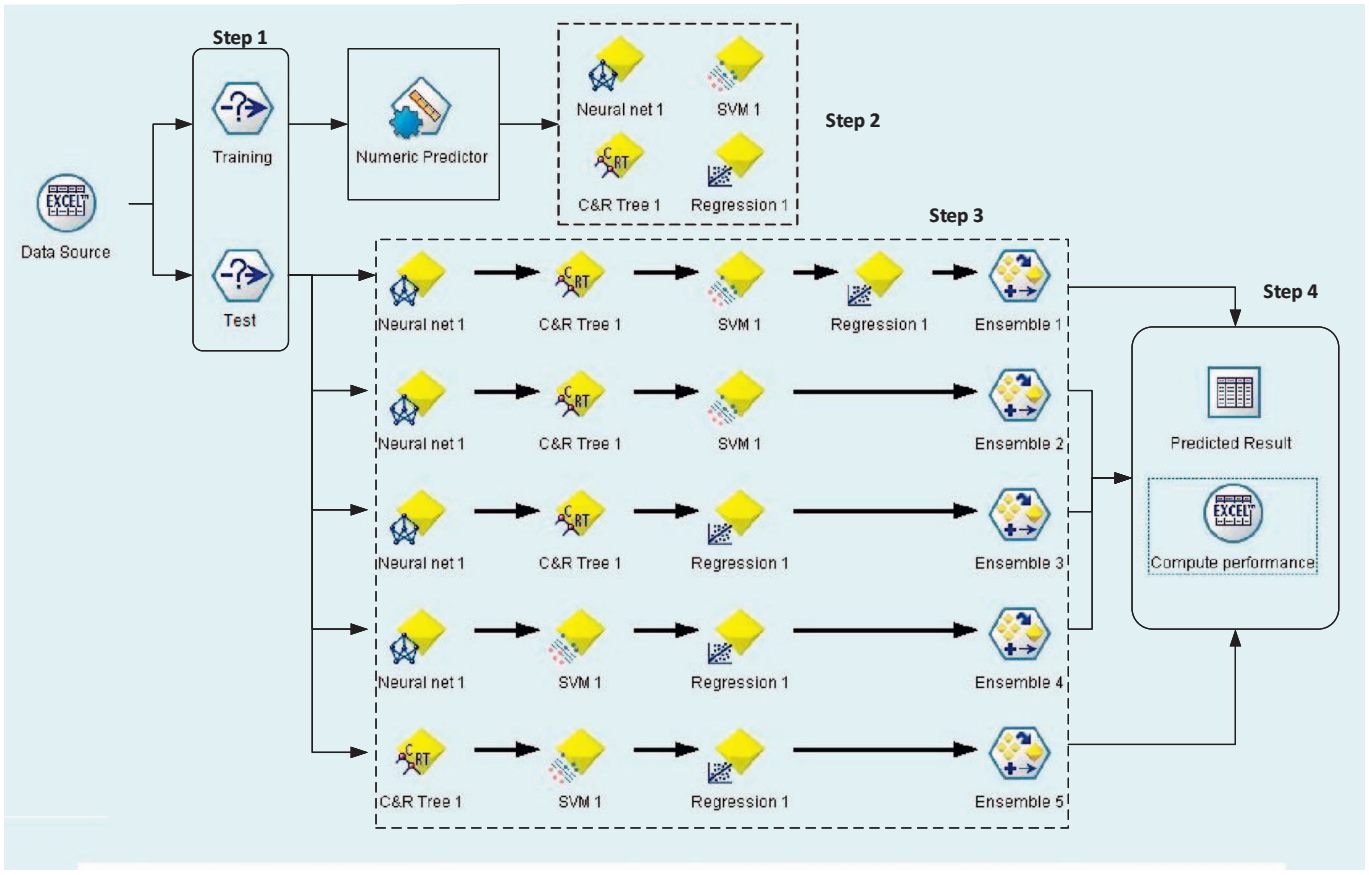


Fig. 7. Stream of voting model.

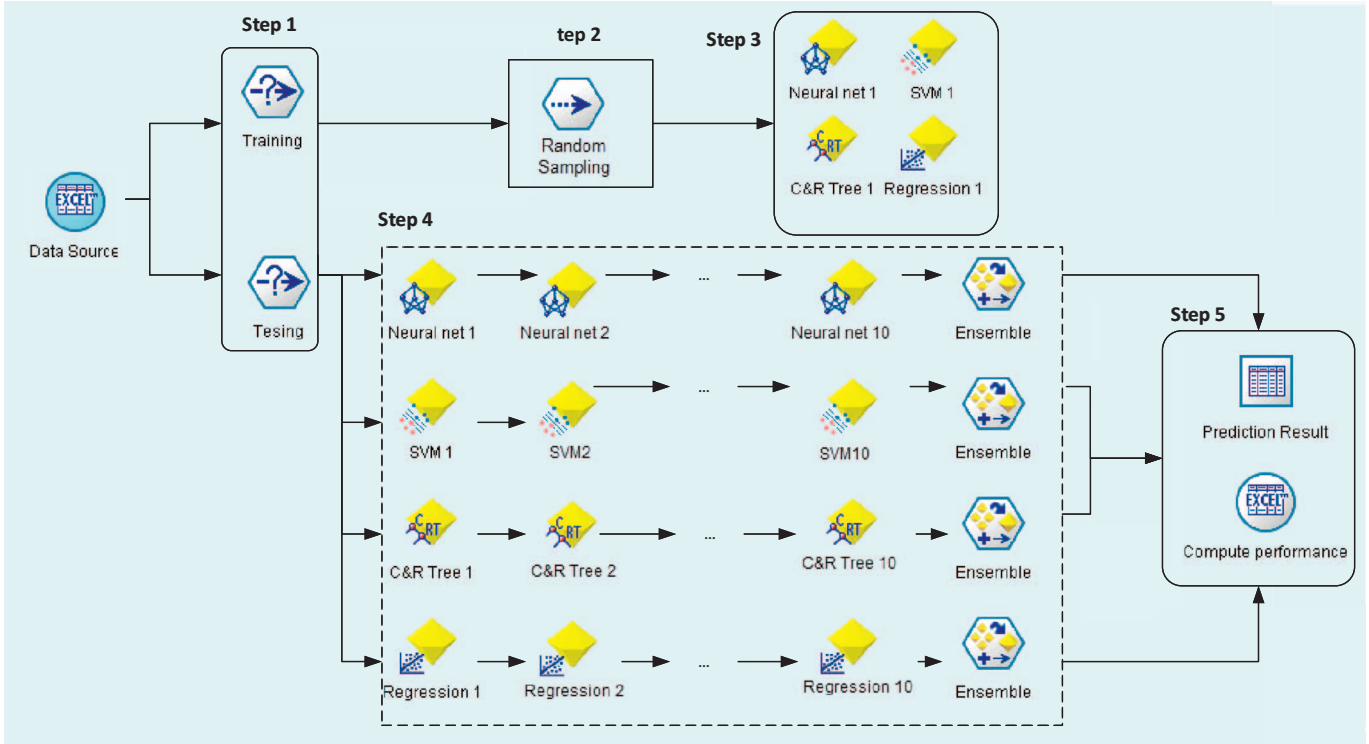


Fig. 8. Stream of bagging model.

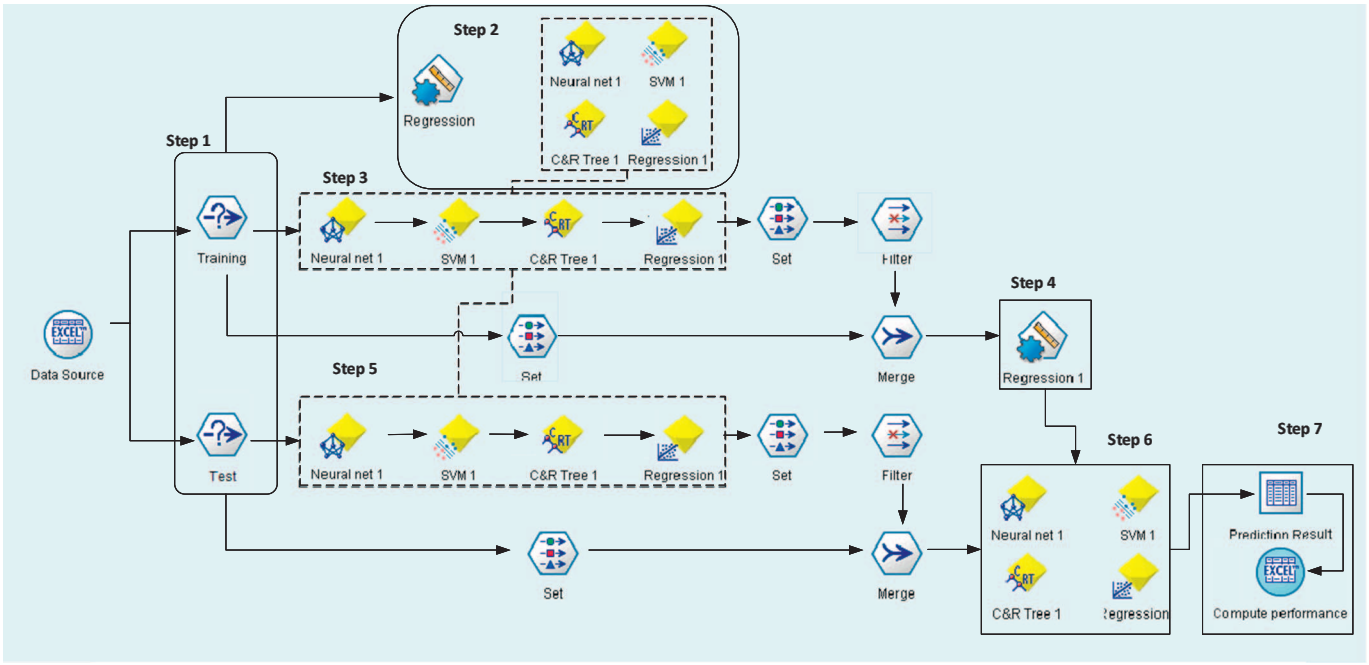


Fig. 9. Stream of stacking model.

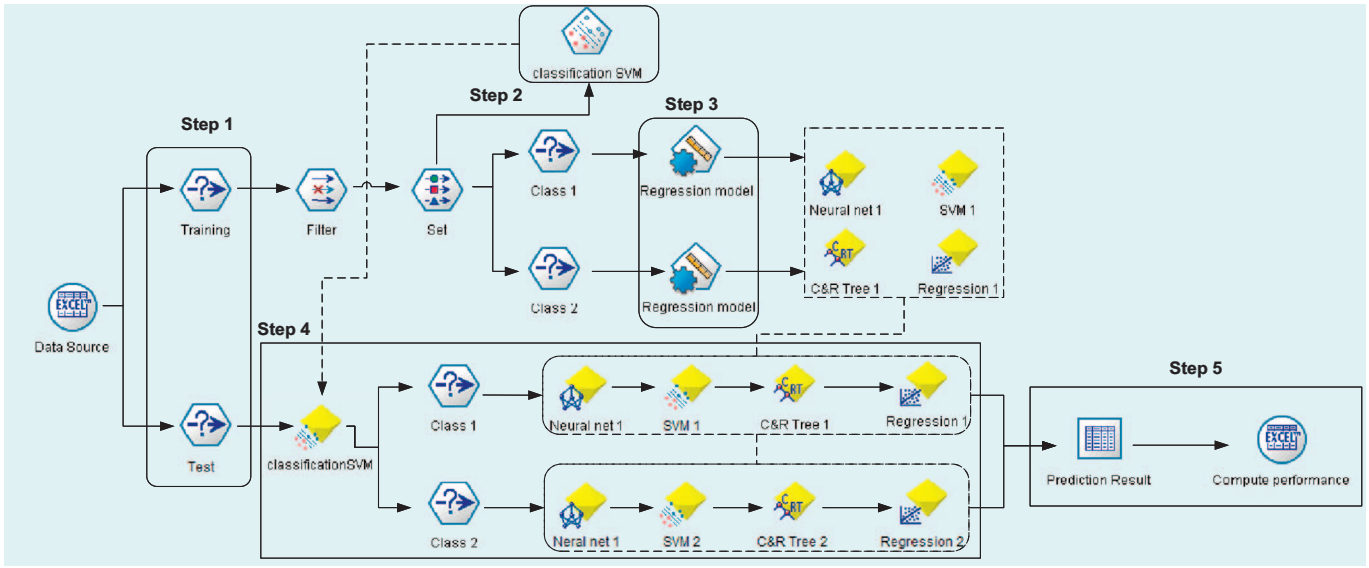


Fig. 10. Stream of tiering model.

(Figs. 7–10) present four ensemble streams.

5.3. Hybrid model

The hybrid metaheuristic regression model was built in MATLAB. MATLAB, with its various functions and toolboxes, is a very useful analytic platform that provides a convenient environment for programming in its M-code language. A graphical user interface (GUI) is a graphical display in one or more windows that contains controls, called components, that allow a user to perform interactive tasks directly. A GUI is designed within the graphical user interface design environment and then implemented by the MATLAB compiler. It can perform any computation, including the reading and writing of data files, communication with other GUIs, and the display of data as tables or plots. The practical efficiency is increased because the user does not need a detailed understanding of the way in which the tasks are performed. Fig. 11 displays the system architecture, abstracted from the implementation.

6. Experimental results

6.1. Baseline scenario

In the first stage of the analytical process, baseline models are implemented in the four software packages - RapidMiner Studio, Microsoft Azure Machine Learning Studio, IBM SPSS Modeler, and WEKA. These packages use four machine learning techniques to establish individual models. The SVM technique, which is not available in Microsoft Azure Machine Learning Studio, is not used. Table 6 presents the predictions thus made.

A comparison confirms that W-REPTree (*i.e.*, Decision Tree) is superior to the other techniques when RapidMiner is used to make

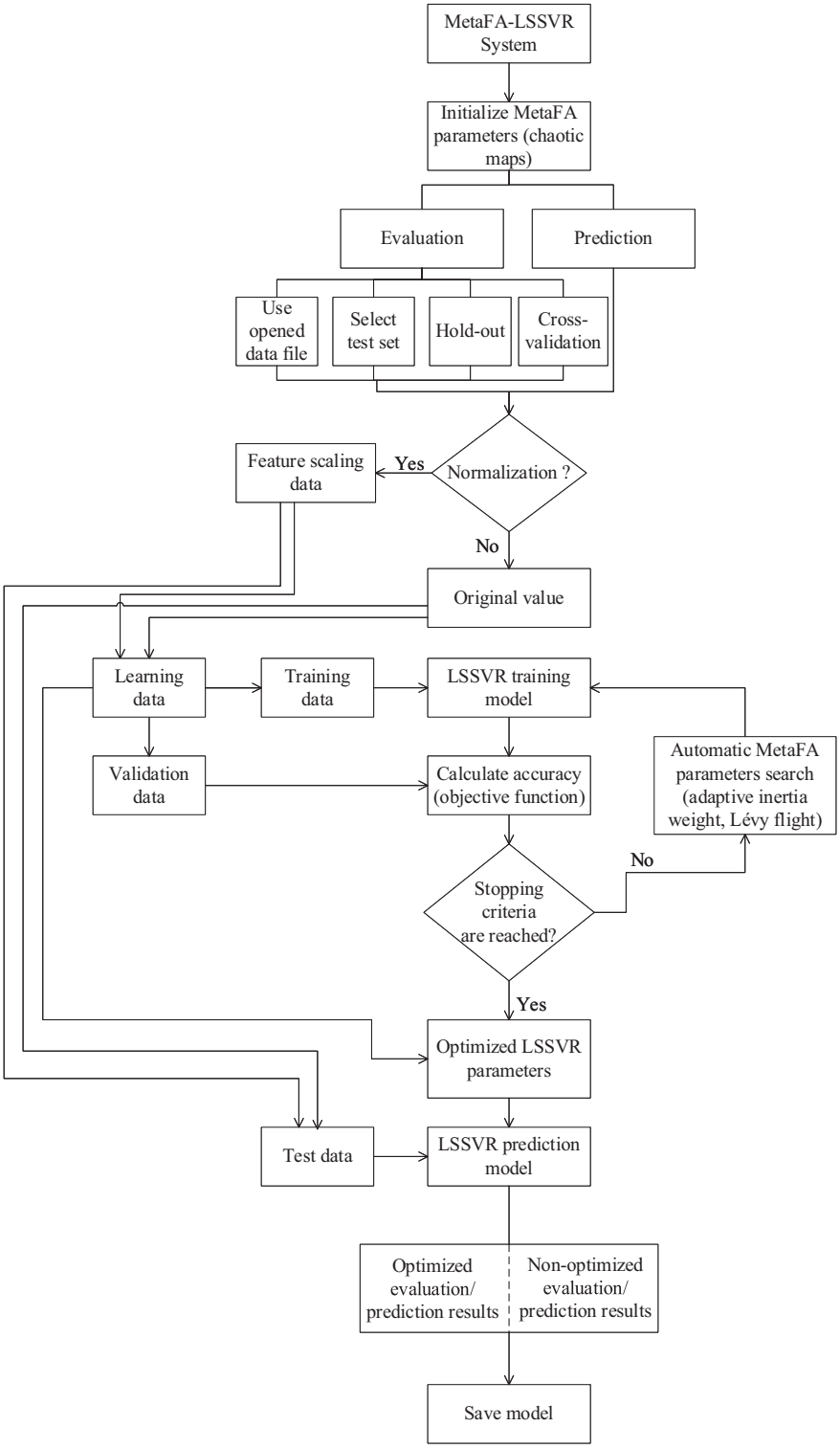


Fig. 11. MetaFA-LSSVR system architecture.

predictions. For instance, W-REPTree not only yielded the lowest values of the three evaluation metrics, RMSE (5.014), MAE (3.890), and MAPE (8.331%); it also yielded the highest R value (0.846). In the WEKA software package, REPTree was the most powerful technique with the three best values of metrics: R (0.846), RMSE (5.006), and MAE (3.912), and the second best value of MAPE (8.364%) indicating that REPTree outperformed the other techniques in this software package.

Artificial Neural Network is the most powerful single model for making predictions in this dataset. In Microsoft Azure Machine Learning Studio and IBM SPSS Modeler, it yielded the best values of R, RMSE, MAE, and MAPE. ANNs outperformed the other techniques in baseline scenario with the Neural Net model in IBM SPSS Modeler, with favorable values of R (0.865), RMSE (4.644), MAE (3.622), and MAPE (7.721%).

A comprehensive comparison among the four software packages with the use of each technique used in the baseline scenario is crucial to identifying the most effective software package for implementing ensemble methods. Table 7 presents the prediction performance of the

Table 6
Prediction result of single models.

Software	Model	Performance measure				
		R	RMSE	MAE	MAPE (%)	$SI_{overall}$
RapidMiner Studio	Neural Net	0.799	5.867	4.468	9.446	0.384 (13)
	SVM	0.556	8.368	5.538	10.815	1.000 (15)
	W-REPTree	0.846	5.014	3.890	8.331	0.124 (4)
	Linear Regression	0.839	5.103	3.926	8.343	0.142 (7)
Microsoft Azure Machine Learning Studio	Neural Net	0.864	4.755	3.717	7.881	0.033 (2)
	Decision Forest Regression	0.799	5.869	4.601	9.793	0.430 (14)
	Linear Regression	0.839	5.095	3.934	8.356	0.143 (8)
WEKA	Multilayer Perceptron	0.804	5.865	4.425	9.268	0.361 (12)
	SMOreg	0.831	5.242	4.059	8.612	0.197 (11)
	REPTree	0.846	5.006	3.912	8.364	0.129 (5)
	Linear Regression	0.840	5.086	3.919	8.328	0.138 (6)
IBM SPSS Modeler	Neural Net	**0.865**	**4.644**	**3.622**	**7.721**	**0.000** (1)
	SVM	0.840	5.035	3.814	8.090	0.101 (3)
	CART	0.835	5.080	3.991	8.534	0.167 (10)
	Regression	0.835	5.115	3.936	8.351	0.147 (9)

Note: Bold value denotes the best performance for each software; (.) indicates the overall ranking.

four individual models, based on the five measures herein.

As shown in the above table, IBM SPSS Modeler is the best package for implementing ANNs and SVR models in the baseline scenario. In particular, Artificial Neural Network in IBM SPSS Modeler, called Neural Net, outperforms other ANN based techniques provided in the other software packages. Support Vector Machine for Regression (SVR) is not available in Microsoft Azure Machine Learning Studio so this technique was applied in the baseline scenario in only three software packages. In comparison with the other software packages, SVR model in IBM SPSS Modeler yielded the best values of R, RMSE, MAE, MAPE, and SI (0.840, 5.035, 3.814, 8.090% and 0.000, respectively).

RapidMiner Studio was the most effective software for implementing the CART (*i.e.*, Classification and Regression Trees) model. Despite the fact that WEKA had the lowest RMSE (5.006) and highest R (0.846), RapidMiner outperformed overall with an SI value of 0.005. The CART based model in RapidMiner Studio, W-REPTree, yielded R = 0.846, RMSE = 5.014, MAE = 3.890, and MAPE = 8.331%. In WEKA, Linear regression yielded the lowest SI value, and so was the most accurate predictive technique. The values of all of the evaluation metrics (R = 0.840, RMSE = 5.086, MAE = 3.919, and MAPE = 8.328%) were better than those achieved using any of the other software packages.

In summary, IBM SPSS Modeler performed excellently in the implementation of the four tested techniques for predicting the quality of water in reservoirs. It yielded the best values of all five measures for two of the four techniques and its Neural Net technique outperformed all the other individual models. Therefore, IBM SPSS Modeler is the best choice for making predictions using ensemble methods.

6.2. Ensemble scenario

Table 8 shows the results obtained using the ensemble methods of voting, bagging, stacking, and tiering, implemented in IBM SPSS Modeler. Each method combined aforementioned individual techniques to improve predictive performance.

The voting method, CART + SVR + LR ensemble, achieved the best result among five ensemble models, with the best values of all evaluation measures. With respect to the bagging method, although the ANN based ensemble model yielded the second best values of R, MAE, and MAPE (0.837, 3.930, and 8.386%, respectively), it had the best overall predictive accuracy with an RMSE value of 5.011 and an overall SI value of 0.663. The ANNs + CART + LR model was the best of the five stacking models as measured by all five evaluation metrics, with R = 0.862, RMSE = 4.693, MAE = 3.682, MAPE = 7.921%, and SI = 0.444.

The tiering method involved dividing the samples into two classes, depending on the CTSI values, with 272 entries in the high class and 1363 entries in the low class. Accordingly, a weighted average was used to evaluate prediction performance, based on the distribution of water quality data. The average performance measures were calculated as [(272 entries of data in the first class / 1635 entries of data in total) × value of measure in first class] + [(1363 entries of data in the second class / 1635 entries of data in total) × value of measure in second class]. The best tiering performance was achieved using ANNs model, which had the highest value of R (0.718) and the lowest value of each of the other measures (RMSE = 3.941, MAE = 3.131, and MAPE = 6.786%).

Overall, the ANNs tiering model was the most stable and accurate of the ensemble models. Specifically, with its high metrics of accuracy and the lowest overall SI value of 0.164, it was the optimal model for use in the ensemble scenario, in which the tiering method was implemented in IBM SPSS Modeler.

Table 7
Performance comparison of four software packages.

Model	Software	Performance Measure				
		R	RMSE	MAE	MAPE (%)	SI
Artificial Neural Network	RapidMiner Studio	0.799	5.867	4.468	9.446	1.000
	Microsoft Azure ML Studio	0.864	4.755	3.717	7.881	0.076
	WEKA	0.804	5.865	4.425	9.268	0.944
	IBM SPSS Modeler	**0.865**	**4.644**	**3.622**	**7.721**	**0.000**
Support Vector Machine for Regression	RapidMiner Studio	0.556	8.368	5.538	10.815	1.000
	Microsoft Azure ML Studio	N/A	N/A	N/A	N/A	N/A
	WEKA	0.831	5.242	4.059	8.612	0.107
	IBM SPSS Modeler	**0.840**	**5.035**	**3.814**	**8.090**	**0.000**
Classification and Regression Trees	RapidMiner Studio	0.846	5.014	**3.890**	**8.331**	**0.005**
	Microsoft Azure ML Studio	0.799	5.869	4.601	9.793	1.000
	WEKA	**0.846**	**5.006**	3.912	8.364	0.014
	IBM SPSS Modeler	0.835	5.080	3.991	8.534	0.151
Linear Regression	RapidMiner Studio	0.839	5.103	3.926	8.343	0.457
	Microsoft Azure ML Studio	0.839	5.095	3.934	8.356	0.575
	WEKA	**0.840**	**5.086**	**3.919**	**8.328**	**0.000**
	IBM SPSS Modeler	0.835	5.115	3.936	8.351	0.953

Note: Bold value denotes the best performance in each technique.

Table 8
Prediction result of ensemble models.

Model		Performance measure				
		R	RMSE	MAE	MAPE (%)	$SI_{overall}$
Voting	ANNs + CART + SVR + LR	0.856	4.768	3.742	8.001	0.492 (12)
	ANNs + CART + SVR	0.857	4.757	3.734	7.984	0.484 (11)
	ANNs + CART + LR	0.859	4.717	3.697	7.914	0.455 (6)
	ANNs + SVR + LR	0.835	5.048	3.961	8.437	0.688 (16)
	CART + SVR + LR	**0.862**	**4.707**	**3.691**	**7.898**	**0.447** (5)
Bagging	ANNs	0.837	**5.011**	3.930	8.386	**0.663** (13)
	SVM	0.826	5.174	4.072	8.660	0.781 (17)
	CART	**0.839**	5.029	3.952	8.477	0.681 (15)
	LR	0.836	5.071	**3.922**	**8.330**	0.666 (14)
Stacking	ANNs + CART + SVR + LR	0.859	4.733	3.718	8.002	0.475 (9)
	ANNs + CART + SVR	0.860	4.717	3.712	7.969	0.466 (8)
	ANNs + CART + LR	**0.862**	**4.693**	**3.682**	**7.921**	**0.444** (4)
	ANNs + SVR + LR	0.825	5.196	4.047	8.704	0.786 (18)
	CART + SVR + LR	0.859	4.744	3.717	8.010	0.478 (10)
Tiering	ANNs	**0.718**	**3.941**	**3.131**	**6.786**	**0.164** (1)
	SVR	0.680	4.255	3.319	7.160	0.369 (3)
	CART	0.643	4.260	3.376	7.381	0.456 (7)
	LR	0.682	4.104	3.238	7.040	0.299 (2)

Note: Bold value denotes the best performance in each ensemble method; (.) indicates the overall ranking.

Table 9
Performance comparison among the best models in each software.

Scenario	Model	Performance measure				
		R	RMSE	MAE	MAPE (%)	$SI_{overall}$
Baseline	RapidMiner Studio: W-REPTree	0.846	5.014	3.890	8.331	0.775 (7)
	Microsoft Azure ML Studio: Neural Net	0.864	4.755	3.717	7.881	0.561 (6)
	IBM SPSS Modeler: Neural Net	0.865	4.644	3.622	7.721	0.480 (3)
	WEKA: REPTree	0.846	5.006	3.912	8.364	0.785 (8)
Ensemble	Voting: CART + SVR + LR	0.862	4.707	3.691	7.900	0.548 (5)
	Bagging: ANNs	0.837	5.011	3.930	8.390	0.810 (9)
	Stacking: ANNs + CART + LR	0.862	4.693	3.682	7.920	0.546 (4)
	Tiering: ANNs	0.718	**3.941**	**3.131**	**6.786**	**0.250** (1)
Hybrid	MetaFA-LSSVR	**0.875**	4.554	3.488	7.463	0.360 (2)

Note: Bold value denotes the best performance for each ensemble method; (.) indicates the overall ranking.

Table 10
Performance comparison among the best models in each scenario.

Scenario	Best model	Performance measure				
		R	RMSE	MAE	MAPE (%)	SI
Baseline	IBM SPSS Modeler: Neural Net	0.865	4.644	3.622	7.721	0.766 (3)
Ensemble	Tiering method: ANNs	0.718	**3.941**	**3.131**	**6.786**	**0.250** (1)
Hybrid	MetaFA-LSSVR	**0.875**	4.554	3.488	7.463	0.580 (2)

Note: Bold value denotes the best performance for each model; (.) indicates the ranking.

6.3. Hybrid scenario

In the hybrid MetaFA-LSSVR model, LSSVR hyper parameters (C, σ) were optimized by MetaFA with 50 fireflies and a maximum of 25 generations. Table 9 presents the performance measures of the best individual AI in each software package, the best ensemble models, and the hybrid metaheuristic regression model, used to predict quality of water in reservoirs.

6.4. Performance comparison

This study identified the best models in the baseline, ensemble, and hybrid scenarios. The best AI single technique is Neural Net in IBM SPSS Modeler while the best ensemble model is ANN based tiering model in the same software package. Finally, the hybrid scenario, MetaFA-LSSVR model, was compared with the best single ensemble models. Table 10 presents the best performance measures achieved using three techniques, baseline, ensemble, and hybrid, for predicting the water quality index of reservoir. Fig. 12 compares them in terms of R, RMSE, MAE, and MAPE.

The ensemble and hybrid models are more accurate than the baseline models. Since it had the highest values of RMSE, MAE, MAPE and SI, the single ANNs model was ranked last whereas the best model was either the ANNs tiering model or the hybrid MetaFA-LSSVR. In particular, the MetaFA-LSSVR model yielded the highest correlation between the predicted and actual values, with an R value of 0.875, while the ANNs tiering model yielded better predictions with the lowest values of RMSE, MAE, and MAPE (3.941, 3.131, and 6.786, respectively). The tiering method based ANNs model achieved the lowest overall SI value of 0.250, and so was the most accurate model for predicting the quality of water in reservoirs.

7. Conclusion

Water quality is one of the most important considerations in reservoir management owing to its extensive effects on nature, animals, and human beings. Not only are reservoir managers and aquatic scientists concerned with the environmental effects of water quality, civil engineers must also consider water quality to comply with regulations in the construction of new reservoirs, or in making structural and

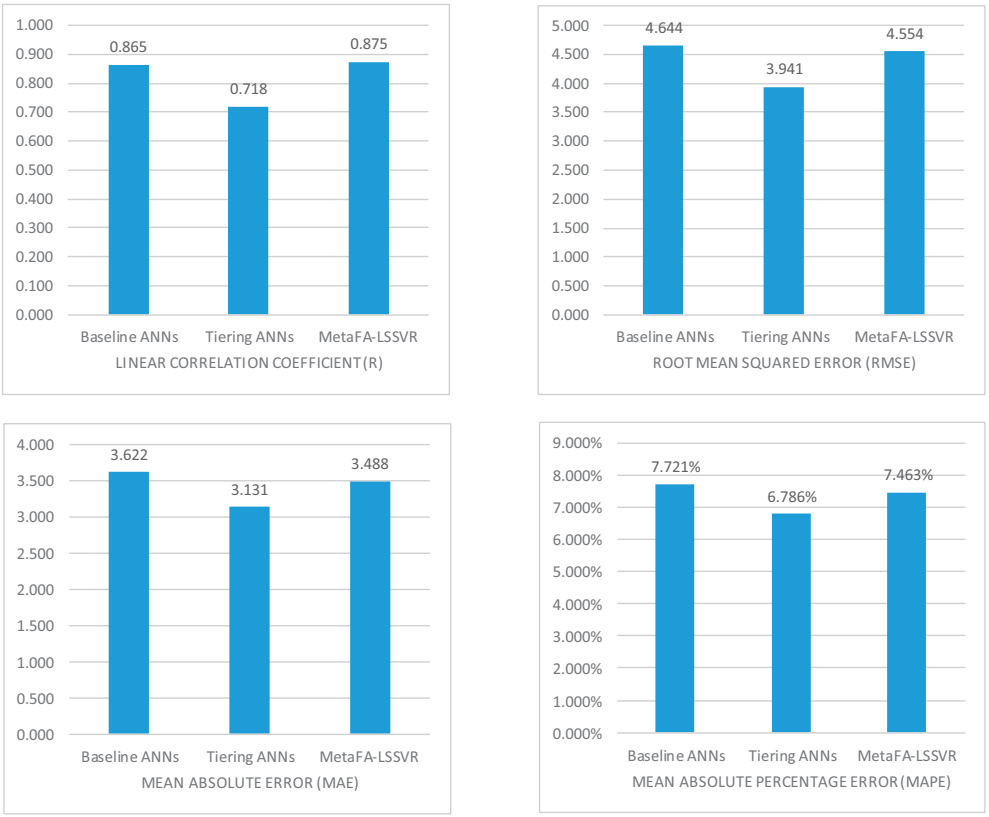


Fig. 12. Comparison of best models in each scenario.

operational modifications to existing reservoirs. Carlson's Trophic State Index (CTSI) has been used all over the world as a metric of the quality of water in reservoirs. This study presented a novel approach to CTSI assessment by integrating artificial intelligence techniques.

To reduce the cost and complexity of conducting chemical experiments, the conventional parameters in the CTSI formula were ignored herein, and a dataset of commonly used water parameters, measured at 20 reservoirs in Taiwan, provided the inputs for the models that were used to predict CTSI. The AI techniques that were implemented in this study are baseline, ensemble and hybrid methods. Four AI techniques (ANNs, SVR, CART, and LR) were applied individually at first and then they were used to generate four groups of ensemble models, involving voting, bagging, stacking, and tiering. Finally, LSSVR was integrated with a metaheuristic firefly optimization algorithm to develop a hybrid AI model.

For diversity of implementation, four analytic platforms - RapidMiner Studio, Microsoft Azure Machine Learning Studio, IBM SPSS Modeler, and WEKA were used to implement the individual AI models. The results thus obtained demonstrated that IBM SPSS Modeler was the most effective in the development of baseline analysis as two out of the four AI techniques performed better on this platform than on any of the others. Additionally, this platform yielded the best performance of all 15 individual models (ANNs), and so is recommended for making predictions of water quality when fast and simple data analytics is required.

The most accurate model among the single ANNs model, the ANNs tiering model, and the MetaFA-LSSVR hybrid model, was identified and used to predict the CTSI of water quality in reservoirs. The baseline ANNs model was the least accurate model, and no ensemble or hybrid model outperformed any other on all measures. MetaFA-LSSVR provided the best fitted linear correlation between actual and predicted CTSI values, with the highest R value of 0.875. The ANNs tiering model, which divides dataset into two classes, was the best overall with the lowest RMSE, MAE, and MAPE (3.941, 3.131, and 6.786%, respectively) and SI = 0.250. This final result revealed that with a large water quality dataset, ensemble tiering was the most powerful method for predicting CTSI. The metaheuristic regression model had the second highest accuracy and was not optimal for such a dataset.

This work provides a new approach of AI models, reducing the complexity of measuring three traditional parameters required in the CTSI formula, as an alternative to the conventional approach to predicting CTSI. This work contributes to the improvement of water quality management by providing a versatile approach that offers diverse predictive techniques to meet the specific requirements of practitioners.

Future work should develop a decision-making strategy for water managers, engineers and other experts who are working with similar databases that considered herein. The tiering method, implemented using IBM SPSS Modeler may be the most accurate in predicting quality of water in reservoirs. Analysts can use the IBM SPSS Modeler software package to make rapid and simple predictions with reasonable accuracy, or use the customized GUI for ease of implementation. However, the GUI will still need a complex hybrid model that integrates metaheuristic optimization and machine learning algorithms to obtain more accurate results. The metaheuristic optimized model is a promising method for water quality assessment.

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